```java
import java.util.ArrayList;
import java.util.Scanner;

class Student {

    int studentId;
    String studentName;
    double studentMarks;

    Student(int studentId, String
studentName, double studentMarks) {
        this.studentId = studentId;
        this.studentName = studentName;
        this.studentMarks = studentMarks;
    }

    public void showStudentDetails() {
        System.out.println("\nStudent ID    : " +
studentId);
        System.out.println("Student Name  : "
+ studentName);
```

```
        System.out.println("Student Marks : "
+ studentMarks);

System.out.println("-----------------------------------")
;
    }
}

public class StudentManagementSystem {

    public static void main(String[] args) {

        Scanner input = new
Scanner(System.in);

        ArrayList<Student> studentList = new
ArrayList<>();

        int option;

        System.out.println("==========
```

```
Welcome to Student Management System
==========");

        do {

            System.out.println("\nChoose an
Option");
            System.out.println("1. Add New
Student");
            System.out.println("2. Display All
Students");
            System.out.println("3. Search
Student by ID");
            System.out.println("4. Remove
Student");
            System.out.println("5. Exit");

            System.out.print("Enter your choice:
");
            option = input.nextInt();
```

```
        switch (option) {

            case 1:

                System.out.print("\nEnter Student ID: ");
                int id = input.nextInt();
                input.nextLine();

                System.out.print("Enter Student Name: ");
                String name = input.nextLine();

                System.out.print("Enter Student Marks: ");
                double marks = input.nextDouble();

                Student newStudent = new Student(id, name, marks);
                studentList.add(newStudent);
```

```
                    System.out.println("\nStudent
added successfully!");
                    break;

                case 2:

                    if (studentList.isEmpty()) {
                        System.out.println("\nNo
student records available.");
                    } else {



System.out.println("\n=========== Student
Records ===========");

                        for (Student student :
studentList) {

student.showStudentDetails();
                        }
```

```
                }

                break;

            case 3:

                System.out.print("\nEnter
Student ID to search: ");
                int searchId = input.nextInt();

                boolean studentFound = false;

                for (Student student :
studentList) {

                    if (student.studentId ==
searchId) {


System.out.println("\nStudent Record
Found");
```

```
student.showStudentDetails();

                              studentFound = true;
                              break;
                          }
                      }

                      if (!studentFound) {

System.out.println("\nStudent not found.");
                      }

                      break;

                  case 4:

                      System.out.print("\nEnter
Student ID to remove: ");
                      int removeId = input.nextInt();
```

```
                    boolean studentRemoved =
false;

                    for (Student student :
studentList) {

                        if (student.studentId ==
removeId) {


studentList.remove(student);



System.out.println("\nStudent removed
successfully!");
                            studentRemoved = true;
                            break;
                        }
                    }

                    if (!studentRemoved) {
```

```
System.out.println("\nStudent not found.");
                    }

                    break;

                case 5:

                    System.out.println("\nThank
you for using Student Management
System!");
                    break;

                default:

                    System.out.println("\nInvalid
choice! Please try again.");
            }

        } while (option != 5);
```

```
        input.close();
    }
}
```